Chapter 16 - TED Audio

TED is the C16/Plus-4 sound-and-video chip. Chapter 6 covers its video half; this chapter covers its audio half. TED audio is a small two-voice generator with a shared noise mode and a master volume - the chip that produced the C16's distinctive bleeps and arpeggios.

16.1 What TED audio can show

Item	Value
Voices	`2`, each a square wave
Frequency range	10-bit divider per voice
Noise	One mode, replaces voice 2
Master volume	`0`–`8` (values `9`–`15` clamp to `8`)
Sound clock	`110,840` Hz PAL / `111,860` Hz NTSC (main clock / 8)

16.2 The register block

The audio block is six bytes wide, at `0xF0F00`–`0xF0F05`. The layout matches the Plus/4's `$FF0E`–`$FF12` exactly, plus the IE-specific TED+ control byte.

Address	Name	Plus/4 alias	Purpose
`0xF0F00`	`TED_FREQ1_LO`	`$FF0E`	Voice 1 frequency low byte.
`0xF0F01`	`TED_FREQ2_LO`	`$FF0F`	Voice 2 frequency low byte.
`0xF0F02`	`TED_FREQ2_HI`	`$FF10`	Voice 2 frequency high bits (low 2 bits).
`0xF0F03`	`TED_SND_CTRL`	`$FF11`	Sound control (volume, voice enables, noise, DAC).
`0xF0F04`	`TED_FREQ1_HI`	`$FF12`	Voice 1 frequency high bits (low 2 bits).
`0xF0F05`	`TED_PLUS_CTRL`	-	TED+ mode enable (bit `0`).

16.2.1 `TED_SND_CTRL` bits

Bit	Field	Meaning
0–3	`VOLUME`	Master volume `0`–`8`; values `9`–`15` clamp to `8`.
4	`SND1ON`	Voice 1 enable.
5	`SND2ON`	Voice 2 enable.
6	`SND2NOISE`	Voice 2 is in noise mode.
7	`SNDDC`	DAC direct-output mode.

16.2.2 Frequency divider

A voice's pitch is set by a 10-bit divider: `FREQn_LO` for the low byte and bits `0`–`1` of `FREQn_HI` for the upper two bits. The output frequency is approximately

```
f = sound_clock / (2 × (1024 − divider))
```

with `sound_clock` = `110,840` Hz on PAL or `111,860` Hz on NTSC.

16.3 The TED+ extension

Bit `0` of `TED_PLUS_CTRL` enables **TED+**: extended volume and DAC modes. With TED+ off, the chip is bit-exact to a Plus/4.

16.4 The TED music player

Address	Name	Purpose
`0xF0F10`	`TED_PLAY_PTR`	Address of the music data.
`0xF0F14`	`TED_PLAY_LEN`	Length in bytes.
`0xF0F18`	`TED_PLAY_CTRL`	Bit `0` = start, bit `1` = stop, bit `2` = loop.
`0xF0F1C`	`TED_PLAY_STATUS`	Bit `0` = busy, bit `1` = error.

The player accepts both the TEDMUSIC (`.tmf`) format and embedded 6502 player binaries.

16.5 BASIC keywords

Form	Effect
`TED TONE`*ch*`,`*divider*	Set voice *ch* (`1`–`2`) frequency.
`TED VOL`*level*	Set master volume (`0`–`8`).
`TED NOISE ON` / `TED NOISE OFF`	Replace voice 2 with the noise generator.
`TED PLUS ON` / `TED PLUS OFF`	Enable / disable TED+.
`TED PLAY`*addr*`[,`*len*`]`	Start music playback.
`TED STOP`	Stop playback.

A BASIC fragment that plays a two-note arpeggio:

```
10 POKE &H000F0800, 1               : REM AUDIO_CTRL = on
20 TED VOL 8
30 POKE &H000F0F03, &H38             : REM SND_CTRL: voices 1+2 on, max volume
40 TED TONE 1, 540                   : REM low note
50 TED TONE 2, 600                   : REM detuned
```

16.6 Putting it together

TED audio is the simplest two-voice engine on this machine. Most programs use it for retro arpeggios or as a tinny background voice. For full Plus/4 tracker music, point `TED_PLAY_PTR/LEN` at the data and write `1` to `TED_PLAY_CTRL`.

The next chapter covers the POKEY, the Atari 8-bit sound chip with four channels and the SAP playback format.